## The laws of exponents and scientific notation

**1** $1 \times 10^{-2} \times 2 \times 10^{-5} = 2 \times 10^{-7}$

**2**

**a** $2.1 \times 10^{8}$ **b** $5.59 \times 10^{7}$ **c** $7.812 \times 10^{21}$ **d** $4.992 \times 10^{33}$

**e** $4.95 \times 10^{4}$ **f** $8.9 \times 10^{10}$ **g** $3.417 \times 10^{10}$ **h** $1.878 \times 10^{-7}$

**i** $8.2 \times 10^{5}$ **j** $2.5 \times 10^{3}$ **k** $2.4 \times 10^{4}$ **l** $4.4 \times 10^{5}$

**m** $7.3 \times 10^{6}$

**3** $1.3690 \times 10^{-7}$

**4** $3.3416 \times 10^{2}$

**5**

**a**

**i** $8.4 \times 10^{4}\ \text{mi}^2$ **ii** $2.1 \times 10^{7}$ people

**b** $4 \times 10^{-3} \dfrac{\text{mi}^2}{\text{person}}$

**6** $3 \times 10^{7}\ \text{s}$

**7** $8 \times 10^{21}$

**8** $1.425 \times 10^{6}\ \text{g}$

**9** $8.179 \times 10^{21}\ \text{kg}$

## Additional questions

**10**

**a** $8 \times 10^{4}$ **b** $5 \times 10^{3}$ **c** $8 \times 10^{-15}$ **d** $1.2 \times 10^{13}$

**e** $1 \times 10^{-9}$ **f** $7.5 \times 10^{-1}$ **g** $2.3 \times 10^{5}$ **h** $1.6 \times 10^{3}$

**i** $2.5 \times 10^{-23}$ **j** $1.2 \times 10^{-12}$ **k** $2 \times 10^{10}$ **l** $4.5 \times 10^{1}$

**m** $3.479 \times 10^{8}$ **n** $2.185 \times 10^{9}$ **o** $3.479 \times 10^{14}$ **p** $1.4 \times 10^{-2}$

**q** $2.25 \times 10^{5}$ **r** $3.92 \times 10^{-2}$

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**11**

**a** $4390$ **b** $0.005\,93$ **c** $0.000\,048$ **d** $0.000\,008\,97$

**12**

**a** $1.35 \times 10^{1}$ **b** $2.257 \times 10^{5}$ **c** $2.25 \times 10^{17}$ **d** $2.25 \times 10^{-1}$

**e** $6.427 \times 10^{6}$ **f** $2.5 \times 10^{-6}$ **g** $5.813 \times 10^{-2}$ **h** $5.025 \times 10^{10}$

**i** $2.9799 \times 10^{2}$ **j** $2.3814 \times 10^{3}$ **k** $5.0087 \times 10^{8}$ **l** $5.1025 \times 10^{-3}$

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